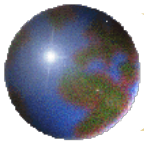
Generalized Wiener Processes

- A Wiener (stochastic) process has a drift rate (i.e. average/mean change per unit of time) of *0* and a variance rate (per unit of time) of *1.0*
- The basic Wiener process, *dz* that has been developed so far has a drift rate of *0* and a variance rate of 1.0.
- In a generalized Wiener process the drift rate and the variance rate can be set equal to any chosen constants



- The drift rate of *0* means that the expected value of *z* at any future time is equal to its current value.
- The variance rate of 1.0 means that the variance of the change in *z* in a time interval of length *T* equals *T*.

Generalized Wiener Processes
(continued)

- A *generalized Wiener process* for a variable *x* can be defined in terms of *dz* as

$$dx = a\,dt + b\,dz$$

- Where,
  - The mean change in *x* per unit time is a
  - Variance of change in x per unit time is b^2

- In a small time interval Δt, the change in the value of x is given by

$$\Delta x = a\ \Delta t + b\ \varepsilon\sqrt{\Delta t}$$

- Thus Δx has a normal distribution with
  - mean of Δx = *a* Δt
  - variance of Δx = b^2 Δt
  - standard deviation of Δx = b $\sqrt{}$Δt
- Similarly the change in the value of x in any time interval *T* is normally distributed with
  - mean of Δx = a*T*
  - variance of Δx = b^2 *T*
  - standard deviation of Δx = b $\sqrt{}$T

Figure 14.1 How a Wiener process is obtained when $\Delta t \to 0$ in equation (14.1).

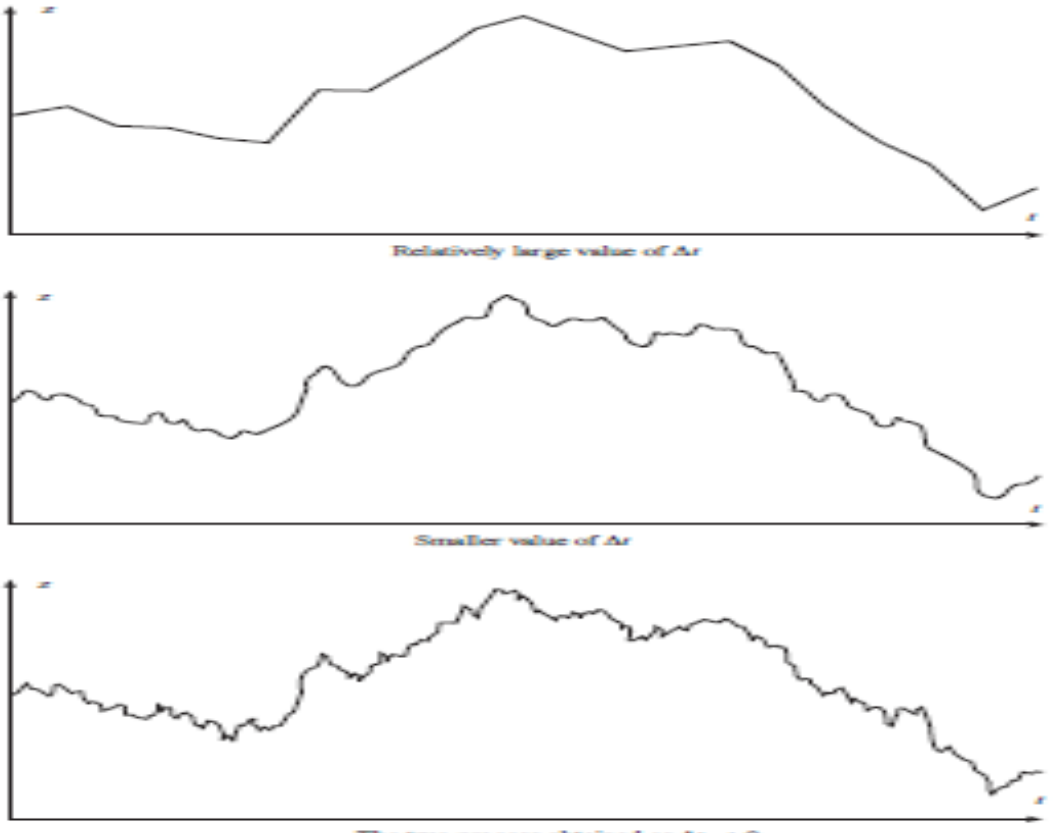
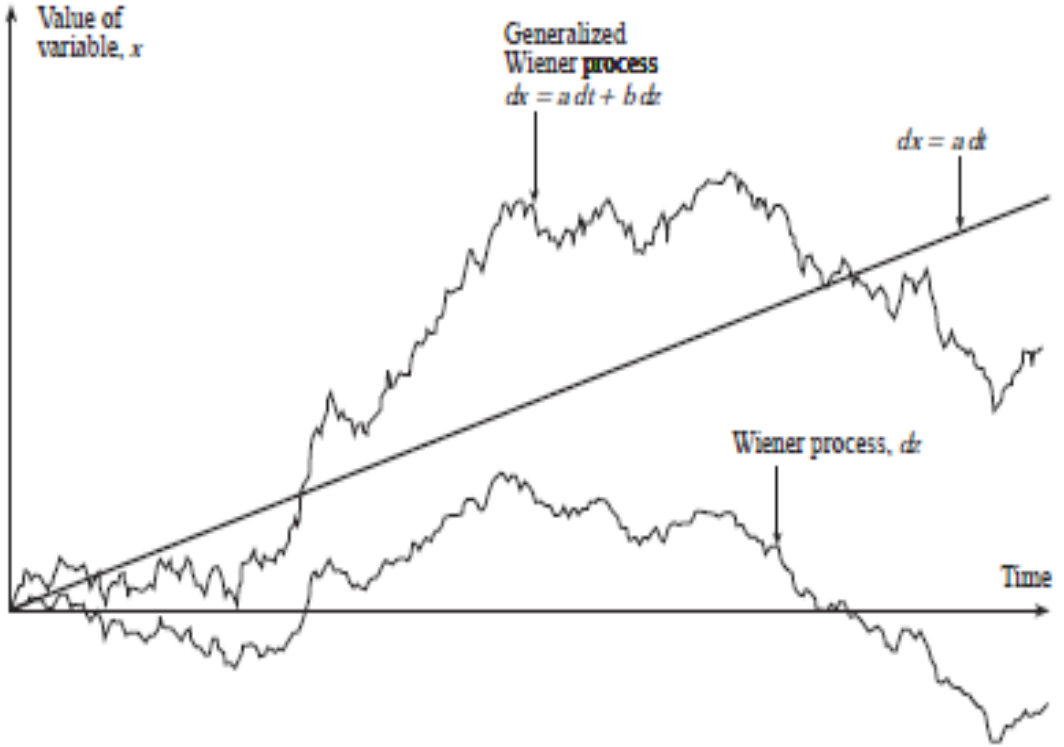


Figure 14.2 Generalized Wiener process with $a = 0.3$ and $b = 1.5$.



Example

- A stock price starts at 40 and has a probability distribution of $\phi(40,100)$ at the end of the year
- If we assume the stochastic process is Markov with no drift, then the process is

$$dS = 10dz$$

- If the stock price were expected to grow by \$8 on average during the year, so that the year-end distribution is $\phi(48,100)$, the process would be

$$dS = 8dt + 10dz$$

Itô Process

- A further type of stochastic process.
- This is a *generalized Wiener process* in which the drift rate and the variance rate are functions of the value of the underlying variable *x* and time *t* written as

$$dx = a(x, t)\,dt + b(x, t)\,dz$$

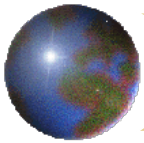
- Both the expected drift rate and variance rate of an Itoˆ process are liable to change over time.
- The discrete time equivalent

$$\Delta x = a(x,t)\Delta t + b(x,t)\varepsilon\sqrt{\Delta t}$$

is only true in the limit as Δt tends to zero.

The Process for a Stock Price

- It is attractive to suggest that a stock price (non-dividend paying stock) follows a generalized Wiener process; that is, it has a constant expected drift rate and a constant variance rate.
- However, this model fails to capture a key aspect of stock prices. That is the expected percentage return required by investors from a stock is independent of the stock's price
- If investors require a 14% per annum expected return when the stock price is $10, then, *ceteris paribus*, they will also require a 14% per annum expected return when it is $50.



- Thus clearly, the assumption of constant expected drift rate is inappropriate and needs to be replaced by the assumption that the expected return is constant
- If S is the stock price at time t, then the expected drift rate in S should be assumed to be μS for some constant parameter μ.
- This means that in a short interval of time,Δt, the expected increase in S is $\mu S \Delta t$.
- The parameter μ is the expected rate of return on the stock.

- If the coefficient of dz (second part of the equation) is zero, so that there is no uncertainty, then this model implies that
  - $\Delta S = \mu S \Delta t$, in the limit as $\Delta t \rightarrow 0$,
  - $dS = \mu S dt$
  - $dS/S = \mu dt$
- Integrating between 0 and T, we get
  - $S_T = S_0 e^{\mu T}$
- This equation shows that when there is no uncertainty, the stock price grows at a continuously compounded rate of μ per unit of time.

- In practice, of course, there is uncertainty.
- Reasonable to assume that the variability of the return in a short period of time is the same regardless of the stock price that is when the stock price is $50 as when it is $10.
- This suggests that the standard deviaton of the change in a short period of time should be proportional to the stock price and leads to the model

$$dS = \mu S\, dt + \sigma S\, dz$$

- Or $ds/s = \mu\, dt + \sigma\, dz$
- The above equation is the most widely used model of stock price behavior and represents the stock price process in the real world.

An Ito Process for Stock Prices

$$dS = \mu S\,dt + \sigma S\,dz$$

where μ is the expected return σ is the volatility.

The process is known as *geometric Brownian motion.*

The discrete time version of the model or its equivalent is

$$\Delta S = \mu S \Delta t + \sigma S \varepsilon \sqrt{\Delta t}$$

- Or
  $\Delta S/S = \mu\Delta t + \sigma\varepsilon\sqrt{\Delta t}$
- It shows that $\Delta S/S$ is normally distributed with mean $\mu\Delta t$ and standard deviation of $\sigma\sqrt{\Delta t}$. In other words,

  $\Delta S/S \sim \Phi(\mu\Delta t, \sigma^2\Delta t)$,

- Example:
- Consider a stock that pays no dividends, has a volatility of 30% per annum, and provides an expected return of 15% per annum with continuous compounding. The process for the stock price is

$$\frac{dS}{S} = 0.15\,dt + 0.30\,dz$$

- Small time interval: $\dfrac{\Delta S}{S} = 0.15\Delta t + 0.30\epsilon\sqrt{\Delta t}$
- For a time interval of 1 week (0.0192 years), it is,

$$\frac{\Delta S}{S} = 0.15 \times 0.0192 + 0.30 \times \sqrt{0.0192}\,\epsilon$$

$$\Delta S = 0.00288S + 0.0416S\epsilon$$

Monte Carlo Simulation

- A Monte Carlo simulation of a stochastic process is a procedure for sampling random outcomes for the process.
- We can sample random paths for the stock price by sampling values for ε
- Suppose as before, $\mu = 0.15$, $\sigma = 0.30$, and Δt = 1 week (=1/52 or 0.0192 years), then

$$\Delta S = 0.15 \times 0.0192 S + 0.30 \times \sqrt{0.0192}\varepsilon$$

or

$$\Delta S = 0.00288 S + 0.0416 S\varepsilon$$

Monte Carlo Simulation – One Sampling Path

Week	Stock Price at Start of Period	Random Sample for ε	Change in Stock Price, ΔS
0	100.00	0.52	2.45
1	102.45	1.44	6.43
2	108.88	−0.86	−3.58
3	105.30	1.46	6.70
4	112.00	−0.69	−2.89
5	109.11	-0.74	-3.04
.	.	.	.
.	.	.	.
10	111.54	2.56	12.20

- From the equation below:

$$\Delta S = 0.00288S + 0.0416S\varepsilon$$

- The change during the first time period is

$$\Delta S = 0.00288 \times 100 + 0.0416 \times 100 \times 0.52 = 2.45$$

- The change during the second time period is

$$\Delta S = 0.00288 \times 102.45 + 0.0416 \times 102.45 \times 1.44 = 6.43$$

- And so on.